



PDF Download
3706598.3714037.pdf
16 January 2026
Total Citations: 1
Total Downloads: 1734

Latest updates: <https://dl.acm.org/doi/10.1145/3706598.3714037>

RESEARCH-ARTICLE

Escape or D13: Understanding Youth Perspectives of AI through Educational Game Co-design

JARED ORDOÑA LIM, Georgia Institute of Technology, Atlanta, GA, United States

GRACE BARKHUFF, Georgia Institute of Technology, Atlanta, GA, United States

JANE AWUAH, Georgia Institute of Technology, Atlanta, GA, United States

SOPHIE CLYDE, Georgia Institute of Technology, Atlanta, GA, United States

RIYA SOGANI, Georgia Institute of Technology, Atlanta, GA, United States

CHRISTINA GARDNER-MCCUNE, University of Florida, Gainesville, FL, United States

[View all](#)

Open Access Support provided by:

[Georgia Institute of Technology](#)

[University of Florida](#)

[Carnegie Mellon University](#)

Published: 26 April 2025

[Citation in BibTeX format](#)

CHI 2025: CHI Conference on Human Factors in Computing Systems
April 26 - May 1, 2025
Yokohama, Japan

Conference Sponsors:
[SIGCHI](#)

Escape or D13: Understanding Youth Perspectives of AI through Educational Game Co-design

Jared Ordoña Lim
School of Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
jlim419@gatech.edu

Grace Barkhuff
School of Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
grace.barkhuff@gatech.edu

Jane Awuah
School of Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
jauuah3@gatech.edu

Sophie Clyde
School of Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
mclcyde3@gatech.edu

Riya Sogani
School of Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
rsogani3@gatech.edu

Christina Gardner-McCune
Department of Computer and
Information Science and Engineering
University of Florida
Gainesville, Florida, USA
gmccune@ufl.edu

David Touretzky
Computer Science Department
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
dst@andrew.cmu.edu

Judith Odili Uchidiuno
School of Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
jiou3@gatech.edu

Abstract

There are many initiatives that teach Artificial Intelligence (AI) literacy to K-12 students. Most downsize college-level instructional materials to grade-level appropriate formats, overlooking students' unique perspectives in the design of curricula. To investigate the use of educational games as a vehicle for uncovering youth's understanding of AI instruction, we co-designed games with 39 Black, Hispanic, and Asian high school girls and non-binary youth to create engaging learning materials for their peers. We conducted qualitative analyses on the designed game artifacts, student discourse, and their feedback on the efficacy of learning activities. This study highlights the benefits of co-design and learning games to uncover students' understanding and ability to apply AI concepts in game-based learning, their emergent perspectives of AI, and the prior knowledge that informs their game design choices. Our research uncovers students' AI misconceptions and informs the design of educational games and grade-level appropriate AI instruction.

CCS Concepts

• **Applied computing** → *Collaborative learning*; E-learning; • **Social and professional topics** → **Computing education**; **K-12 education**.

Keywords

Co-design, game-based learning, AI Education, Youth, AI Literacy



This work is licensed under a Creative Commons Attribution 4.0 International License.
CHI '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1394-1/25/04

<https://doi.org/10.1145/3706598.3714037>

ACM Reference Format:

Jared Ordoña Lim, Grace Barkhuff, Jane Awuah, Sophie Clyde, Riya Sogani, Christina Gardner-McCune, David Touretzky, and Judith Odili Uchidiuno. 2025. Escape or D13: Understanding Youth Perspectives of AI through Educational Game Co-design. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 17 pages. <https://doi.org/10.1145/3706598.3714037>

1 Introduction

There is growing interest in teaching Artificial Intelligence (AI), Computer Vision (CV), and Machine Learning (ML) topics to children around the world [14, 70, 80, 106]. As youth increasingly use ubiquitous AI technologies such as Amazon Alexa and ChatGPT in their everyday lives, scholars have argued for expansions to AI education that teach youth fundamental AI concepts and facilitate deeper critical thinking [8, 23, 24, 52, 62, 91, 93, 102, 105]. Significant amounts of prior work have focused on the development of AI literacy in K-12 through identifying principles of AI education appropriate for primary and secondary students in traditional or formal classroom settings, and offer insights into developing professional development programs for K-12 teachers through culturally relevant and engaging AI curricula [1–4, 91]. In addition to existing professional development resources for teachers, online platforms and tools that supplement teaching AI and Machine Learning (ML) to adolescents have surged in popularity, such as Google's Teachable Machine [37], Code.org's AI For Oceans [16], LearnML [29], and Machine Learning for Kids [53].

Game-based learning (GBL) has increasingly been employed to scaffold AI education to both K-12 and college students [106]. GBL utilizes educational games to stimulate students' interest in the subject matter and reward students for meeting learning objectives. Zhan et al. conducted a systematic literature review of GBL in AI education and found an abundance of research studies focused on

college aged students – however, they observed a lack of similar research depth in K-12 settings [106], which is the population that GBL may benefit the most [47, 55, 103]. In addition, Zhan et al. found that programming occupied the largest portion of game-based course concepts, while concepts, algorithms, application, security, and ethical issues were not addressed enough [106].

Many of these initiatives are successful in stimulating interest in AI and STEM careers, but often assume students to have a background in Computer Science (CS) education and an understanding of CS concepts, or focus on the technical programming aspects of AI [45, 57, 99]. These assumptions reinforce structural barriers for students who want to learn AI but lack formal CS instruction or technical programming backgrounds. Traditional US CS classrooms are also historically overrepresented in White and Asian males [60], which further limits AI education interventions, and thus AI literacy, to a small subset of the youth population.

As a response to the lack of diverse representation from women and racially minoritized identities in computing fields, prior work has explored using culturally responsive teaching to broaden participation in computing education [15, 42]. Culturally responsive teaching requires teachers to understand students' culture, perspectives, and lived experiences as tools for better and more inclusive classroom instruction [34]. In the context of CS education, prior work has explored equity-focused professional development programs for teachers in classroom settings, and noted how CS teachers were able to successfully incorporate students' culture into their instructional design, resulting in the empowerment of both teachers and students [15, 78].

Participatory design (PD) has also been used as a method to integrate student's voices to develop culturally responsive computing education [18]. One specific approach within PD is co-design, where design experts work closely with end users (youth in this context) to co-create technology together, with the ultimate goal of producing more informed and equitable end products and learning outcomes [28]. One example of successful culturally responsive design through co-design partnerships was by Chen et al., who involved Black and mixed-race youth into the design process of a video game. The young participants argued for seeing more identity representation of themselves and felt empowered when they were given the tools to customize their characters to their physical identity [12]. These insights suggest that designing AI curricula and instruction targeted for youth should be done in collaboration with both instructors and youth to represent youth's own unique perspectives.

Our research addresses these gaps in GBL for AI education in K-12 settings, as well as issues in inclusive AI education for K-12 students from diverse backgrounds. In our study, we engage with predominantly high school girls and non-binary youth of color, who are historically underrepresented in the computing field. We incorporated their voices into AI education by teaching them basic AI concepts, followed by co-designing games with them to uncover students' understanding of AI and maximize their engagement. Unlike prior work where GBL AI approaches depend on students' technical programming skills [106], we mostly focus on the fundamental AI concepts and algorithms through unplugged activities and minimal programming activities.

We investigate the following research questions:

- (1) What do co-designed games reveal about students' understanding and ability to apply fundamental AI concepts?
- (2) What does co-design reveal about students' perceptions of the role of AI in educational games?
- (3) What prior knowledge do students draw upon for AI-related educational game design?

Our contributions to the HCI community are twofold: we demonstrate the benefits of co-design and educational games as vehicles for uncovering students' emergent perspectives and misconceptions about AI, and present findings that improve AI curriculum and educational game design for students from diverse backgrounds.

2 Prior Work

2.1 K-12 AI Education

The availability and spread of AI education for K-12 has significantly increased in the last decade as AI technologies have expanded in global recognition [62, 99]. Most champions for implementing AI education in K-12 highlight the benefits to science and industry [100, 105]. For example, Wang et al. note that learning AI can help motivate students and stimulate creativity by demonstrating the power of AI applications [100], but only if taught effectively. Many studies have examined pedagogical design and effectiveness of AI education for K-12 programs, but as noted by Yue et al.'s systematic literature review of AI education programs, most of these studies have small sample sizes and the programs lack a standard curriculum [105].

In 2019, Touretzky et al. [91] partnered with various AI experts to create a list of five "big ideas," which covered fundamental topics relevant to K-12 AI education. They developed a series of grade band progression charts that lay out the concepts and subconcepts of the Five Big Ideas and how they should be conveyed at each of four grade bands (K-2, 3-5, 6-8, and 9-12) – these big ideas have influenced many curricula designed for K-12 AI education [1], such as the *AI4GA* program [2].

The Five Big Ideas are as follows:

- (1) *Perception*: Computers perceive the world using sensors.
- (2) *Representation and Reasoning*: Agents maintain representations of the world and use them for reasoning.
- (3) *Learning*: Computers can learn from data.
- (4) *Natural Interaction*: Intelligent agents require many kinds of information to interact naturally with humans.
- (5) *Societal Impact*: AI can impact society in both positive and negative ways.

While these big ideas define clear guidelines on what to teach students in K-12 about AI, researchers have debated on the best format to relay these concepts to students. Wong et al. designed a curriculum for Hong Kong students that split these concepts by grade level, resulting in students gaining more computational thinking (CT) & programming skills as they advanced through each grade [102]. Many AI curricula likewise focus on computational skills, utilizing various tools and software packages to introduce students to AI via hands-on activities and project-based learning that facilitate greater understanding of AI concepts [57]. Some popular tools used included machine learning demonstrations such as Google's Teachable Machine, AI extensions to the block-based

programming software Scratch, programming languages such as Python, and robotics focused software like Popbot or DUPLO block programming to make LEGO robots [104]. These examples show instructor-led curriculum design in AI education. Our research extends this prior work on AI curricula by providing students' perspectives on these AI concepts through student-led, creative co-design sessions to allow them to express their understanding of AI concepts and applications.

2.2 Game-Based Learning (GBL) in K-12 AI Education

GBL has been used in K-12 contexts in a variety of subject matters to enhance learning outcomes by supporting students' learning & problem solving skills [21]. Prior work has identified the many educational benefits that digital game-based learning can yield for students. Educational games are designed to draw out students' intrinsic motivations, interests, and willingness to learn the subject matter at hand [21, 106]. As a result, these games tend to boost students' cognitive and social-emotional growth, enhance their digital skills, and foster autonomy, critical thinking, and creativity [47, 55]. However, GBL's effectiveness is dependent on various factors including students' learning environment, cultural background, and self-efficacy in the subject matter [31, 81]. Solyst, et al. found a significant variation of student learning needs from GBL that depended on learning resources available to them, their learning goals, and personal demographics [83]. When done correctly, co-designing educational games can be an effective way to support the varying needs of students, vulnerable student populations, and overall improve student self-efficacy. The process of game co-design with students has also been shown to develop creativity, strengthen bonds amongst students, accomplish learning goals, and more [6, 43, 54]. In the past, educational game co-design has been done with underrepresented student populations, including Black youth [12], those from lower socio-economic backgrounds [96], those who are neurodiverse [11], and low-vision [65] populations. These studies indicate how co-designed educational games have the potential for empowerment of underrepresented student populations, and can increase student self-efficacy against stereotype threat [87]. Building on this, our research aims to extend AI curricula using co-design participatory approaches to involve underrepresented students for insights into building equitable AI education curricula.

Scholars have presented different critiques of GBL initiatives. Nicholson argued that while reward-based gamification tended to be effective for achieving short-term goals, relying on rewards may undermine intrinsic motivation [66]. Furthermore, some GBL mechanics such as leaderboards or badge systems can potentially lead to decreased motivation and satisfaction over time [33, 40]. Another criticism of GBL is how video game culture, from which GBL design originates, is often characterized by sexism and marginalization of women [19, 33]. Paaßen et al.'s literature review on the association of masculinity and video game play found many studies suggesting men are more likely to identify as "gamers" in both professional and private settings, whereas feminine identity can be incongruent with a "gamer" identity. They also noted that women gamers were much more likely to face harassment, sexism, exclusion, and marginalization in video game environments and development [68].

Studies have also found gender disparities in game genres, with "hardcore" game genres, such as shooters or strategy games, being more male-dominated, while "casual" game genres were comparatively more gender-equal [10, 97]. Regarding students' perception of the learning value of GBL, Jossan et al. identified that men had a more positive view of GBL than women did [49].

In response, some researchers have pioneered female-oriented game design or "girl games" for instructional design [22], although others have criticized "girl games" in general as reaffirming traditional gender roles and stereotypes [22]. Kafai's literature review on game designs in learning sciences indicates that games with more in-game choices and personalization, and features for players to personalize the assets of games, were successful in appealing and engaging for both girls and boys [50]. In our study, we engage with girls and non-binary youth and contribute to prior research investigating the design preferences of underrepresented gamers, how they adapt AI curricula, and how GBL can support learning for their peers.

2.3 Co-design for AI Education

Cooperative inquiry and co-design with youth is becoming more common as a way to promote inclusivity and lend power to youth as equal partners in learning, whether it be learning STEM, design thinking, or social and emotional learning [46, 69, 71, 76, 92]. Prior work by Druin identifies *cooperative inquiry* as a PD research approach which empowers youth as equal partners in technological development, and supports intergenerational design teams in developing youth-centric technology [27]. Cooperative inquiry is grounded in three critical aspects as identified by Druin: *multidisciplinary research partnership*, *field research*, and *iterative low-tech and high-tech prototyping* [27].

PD has also been used in STEM and computing education contexts to integrate stakeholders' voices and develop culturally responsive computing curricula [4, 17, 18]. Prior work explored partnerships between AI experts, teachers, and policymakers to provide professional development for teachers to expand AI literacy for K-12 students [91]. For example, Touretzky et al. collaborated with computer science teachers to reimagine the five big ideas in AI and co-design instructional materials for middle school students [91]. Dai investigated teachers' construction of teaching practices through ethnography and active co-design sessions with researchers and identified three patterns in teacher-scientist negotiation: reciprocal negotiation of knowledge, observation and interpretation of scientist practice, and inconsistency with knowledge translation [20]. Similarly to Touretzky et al., Roschelle et al. convened with AI experts and learning experts to address issues of ethics, equity, and potential harm in AI education, where they argue for deeper partnerships between policymakers, practitioners, and researchers [77]. These examples highlight current efforts to adapt AI literacy concepts to younger students in formal classroom interventions by using participatory design with various stakeholders. Our research further extends these studies by showing the benefits of co-design to uncover students' understanding of AI instruction, and incorporating youth's voices into designing culturally responsive AI educational games.

2.4 Youth Perspectives of AI and AI Fairness

Students' backgrounds and exposure to AI agents can lead to misconceptions or a lack of critical understanding about how AI works [52]. For example, a common misconception about AI agents such as Amazon's Alexa is that they have a phone inside them that is used to search for answers and then speak them out loud [8, 25, 26]. Sociocultural factors can influence youth's perceptions of AI. For example, Druga et al. found that students from outside the United States were more critical of AI compared to those from the United States, and that higher socioeconomic status (SES) students were more knowledgeable about AI than their lower SES peers [24]. In our research, we engaged with girls and non-binary youth of color, who are historically underrepresented in computing fields [60], and we contribute to the understanding of underrepresented youths' perceptions and critical perspectives on AI applications.

Research also shows that youth, when prompted, are able to critically reflect on the impacts of AI, although they often lack vocabulary or terminology for deeper discourse [26, 56, 79, 95]. Prior work by Arsenio et al. identified that youth are acutely aware of institutional fairness without direct instruction [7]. With this awareness, Vakil identified that youth utilize their own situated identities and cultures to critique everyday technologies such as AI [95]. Youth's exposure to AI content knowledge lead to them becoming more aware and critical of the abilities of AI, and they increasingly attribute the agency of AI systems towards system designers rather than AI itself [23, 26]. To further investigate youth's perspectives of AI and algorithmic fairness, Skinner et al. conducted exploratory co-design sessions with youth of color to understand their perception of fairness, where they found that facilitating role-playing and tabletop polling to have collaborative discussions around algorithmic justice was successful in stimulating open discussions among youth of color [82]. Building on this, Druga found that youth used the scientific method to deeply reflect on the agency of AI in the context of AI voice assistants [23]. Our study contributes to understanding how youth can critically reflect through GBL as a framework to foster deeper critical thinking around the societal impacts of AI.

In addition, research studies show youths' perceptions of AI often include assigning human characteristics to AI agents. For example, in a study of middle school youth exploring the use of ChatGPT, one of the ways they interacted with the agent was through socializing as if they were talking to another person, and argue that leveraging human-like qualities assigned to AI agents, in addition to creativity and the ability for youth to customize and co-create their AI agent, can make "personally meaningful" learning experiences for youth about AI [8]. Similarly, multiple studies found that youth use human qualities such as "smartness" to describe AI's capabilities. Sometimes this is prompted by the researchers directly asking how smart youth think the AI is [25, 101], but other times their perception of AI as "smart" is unprompted [8, 52]. One research study prompted young children ages 3-10 to decide if a series of AI agents were smarter, neutral, or not as smart as them. The older children felt that the agents were generally smarter than them whereas the younger children responded with varied levels of agreement [25]. Williams et al. found that youth with less AI knowledge were more likely to think the AI agent was more like

a toy whereas the children with more AI knowledge were more likely to think the AI agent was more like a person [101]. Our research extends this body of work by investigating how youth's humanizing of AI agents impacts educational game design, and their understanding of AI ethics and fairness in society.

3 Methodology

3.1 Context

This study was conducted in collaboration with a STEM internship program for high school girls and nonbinary students hosted at an R1 university in the Southeastern US. Students apply to the internship program, which works to connect young women and gender minorities to new opportunities through computing, and are selected for participation. It is a year-long high-school-to-college partnership. During the internship program, different faculty from the R1 university teach elective short courses to the students throughout the year. Our short course introduced students to fundamental AI concepts using materials from a publicly available 9-week elective middle school AI course adapted from Touretzky et al.'s Five Big Ideas [91]. This course was co-designed with teachers from 9 local school districts, and teachers from several Southeastern US states have taught it to over 1500 middle school students in diverse cultural and economic contexts. In our study, the short course was led by a research professor with expertise in K-12 AI literacy, participatory design, learning sciences, and educational technologies with help from six graduate and undergraduate research team members, who served as co-design facilitators. The research professor also coordinated the co-design sessions with all the teachers who designed the original curriculum.

3.2 Student Demographics

Thirty-nine high school girls and non-binary youth from underrepresented computing, racial, and socioeconomic backgrounds were enrolled in our elective course; however, four dropped before our last session. Students' self-identification in the initial survey showed that 51% of the initial applicants were in grade 11, while 22% and 27% were in grades 10 and 12, respectively. 93% indicated that they had taken some CS-related classes, and 16% had English as their second language. We did not notice any difficulty in their ability to collaborate and participate in the co-design sessions due to these factors. Twenty-four students identified as Black/African-American, nine as Asian, and two as Hispanic/Latinx – the rest self-identified as mixed-race or preferred not to state. Relating to gender, thirty-six identified as women, two as nonbinary, and one preferred not to state.

3.3 Positionality

Given that our research participants were primarily girls of color, some of the co-designed games center narratives that may contain racial and gender biases in computing and participatory design; therefore, we include a positionality statement indicative of the makeup and experience of our research team.

We are a diverse team of professionals with experience as HCI and design researchers, learning scientists, K-12 STEM instructors, AI education curriculum developers, game developers, and software engineers. We come from diverse cultural backgrounds

Week (Topic)	Learning Objectives & Activities
Week 1 (Intro)	Overview of 5 Big AI Ideas [2, 91]; Student team formation
Week 2 (Perception)	Overview of Computer Vision (CV) topics, including Light Detection & Radar (LIDAR) and Radio Detection & Ranging (RADAR) Sensors; Overview of image filtering technologies; understanding how computers detect objects
Week 3 (Representation & Reasoning)	Overview of graph representations and how algorithms use them. How Global Positioning System (GPS) works; Understanding how Breadth-First Search (BFS) and Depth-First Search (DFS) works
Week 4 (Learning)	Overview of how computers learn from data. Four types of reasoning algorithms: classification, prediction, recommendation, and planning/scheduling; How supervised machine learning models work, and how to train binary classifiers
Week 5 (Natural Language Processing)	How computers interpret sound and recognize human speech; Semantic differences between groups or clusters of words
Week 6 (AI Ethics)	Reflect on the best use cases of AI in your everyday lives

Table 1: Overview of Class Structure

including Black, Hispanic, White, and Indian women, and Asian-American and White men. In prior work, we have co-designed computer programming games with K-12 boys and girls from low- and middle-income communities, rural and urban areas, and have published research on how different socio-cultural factors impact the co-design process and game output. We have also partnered with K-12 public school teachers in the US and AI experts to develop AI literacy curricula for K-12 students.

We recognize that our lived experiences may differ from our participants due to age, generational context, and cultural upbringing, and cannot claim to fully embody the positionalities of our students. We do not claim that our insights generalize to all youth from underrepresented communities. Rather, we focus on presenting nuanced reflections of the co-design process, emphasizing the emergent perspectives of the students involved in our study.

3.4 Class Curriculum & Structure

This study took place over a period of five weeks, with two meeting days a week. The first session of the week focused on instructing students on an AI topic by the lead instructor, a co-design session by students and researchers to create games that teach the AI topic, and then presentations of each design. In the second session of the week, participants played selected AI education and literacy games related to the topic they learned in the previous session and critiqued each game’s efficacy for teaching AI concepts. We used the AI4GA curricula as teaching materials in our study, which was co-designed with K-12 teachers and been used in over 9 school districts and 1,500 students across the US state of Georgia [2]: a short description of each module is described in Table 1.

In the *Perception*, *Representation & Reasoning* and *Learning* modules, we asked students to co-design an activity or game to teach the topics they learned within each module. In the *NLP* and *AI Ethics* modules, we included board games as an option for students, to reduce the limitations imposed by the computer game format, and allow for more creative expression.

3.5 Co-Design teams

Each co-design team was made up of one research team member and 6-8 students. On the first day, we asked students to identify themselves as artists, gamers, teachers, and storytellers. We subsequently created teams of students such that each team would have at least one of each role. Seven teams were created after the first session, and each team’s composition remained the same throughout the study. Each team was comprised of predominantly Black girls. We did not collect information about students’ prior gaming experiences, nor teach them how to design games. These co-design sessions were 30 minutes long and framed as design challenges, where we asked teams to create a fun game to teach students topics related to what they learned in the day’s lesson.

Each team received a poster board, a poster-sized sheets of paper, construction paper in different colors, markers, colored pens and pencils, modeling clay (Play-Doh), sticky notes, and pipe cleaners. After each co-design session, teams presented their game ideas to the whole class, where they showed their game and described how the game worked, the number of players needed to play, and the game’s objective. The lead instructor sometimes prompted clarifying questions to ask students to elaborate on the game’s description and other students sometimes asked questions or provided feedback to each game.

All the co-design sessions took place in a large classroom. The classroom had 3 cameras that captured visual footage of co-design teams from different angles, and each team’s co-design interactions were captured with an audio recorder. Additionally, we took photographs of participants and their design artifacts during design sessions. After each meeting, the research team held a debrief meeting to discuss the co-design activity, collected artifacts, and planned activities for upcoming sessions. These debrief meetings were also recorded, and later transcribed for analyses. We received informed consent from students and their families, and our research was approved by our institution’s Institutional Review Board (IRB). Students received compensation for participating in the internship program but not directly from our research study.

3.6 Data Analysis

We used an inductive data analysis approach and followed thematic analysis [88] to analyze the students' game presentations and co-design recordings. We annotated each co-design sessions' video and audio recordings as they related to our research questions. These annotations included how AI was conceptualized in the games, when new ideas were introduced during co-design sessions, where these ideas came from, the genres or themes of each game, and the objective of each game. We discussed each game and all annotations extensively, and clarified points of confusion. We documented disputed annotations and had open discussions as a research team, following Hammer and Berland's position on qualitative codes as outcomes of the data analysis process [39]. Where there were disagreements, each member of the team separately reviewed the game presentation recordings, then we triangulated independent findings with the co-designed game artifacts until we reached agreement.

4 Findings

We present our results organized by themes summarizing the ways that students demonstrated and applied fundamental AI concepts from their co-design sessions (RQ1), what the co-design sessions revealed about their emerging perspectives of AI (RQ2), and the prior knowledge employed when designing AI-related educational games (RQ3). We opt for *thick descriptions* [35, 72] to present the co-design process between students and adult facilitators. To preserve anonymity, quotes and images are attributed to the team who designed that game rather than individuals.

4.1 RQ1: What do co-designed games reveal about students' understanding and ability to apply fundamental AI concepts?

From the co-designed artifacts, students constructed game narratives that envisioned many different ways AI technologies can be used. Students demonstrated that they can go far beyond the AI concepts we taught them, and in the games, students used storytelling to imagine scenarios of where AI applications can be useful. In the subsections below, we present the insights gained across different AI literacy topics.

4.1.1 Students demonstrated an understanding that AI applications can be used in a variety of real-world or imaginary scenarios and used them to contextualize games' learning outcomes. In many co-designed games, students showcased their creativity by constructing elaborate in-game narratives that demonstrated different AI applications. Oftentimes, students utilized roleplaying to teach players about different applications of AI and its possible limitations.

For example, during the *Perception* and *Learning* modules, students primarily developed puzzle-solving and roleplaying games that contextualized learning outcomes by asking players to take on a role within a broader game narrative. The games students developed to teach *Perception* were very diverse, ranging from narratives about racing to open-world exploration. To incorporate what they learned, students included LIDAR, RADAR, and self-driving cars as tools for exploration in their game designs. Students also co-designed puzzle solving and roleplaying games in the *Learning*

module. The games were diverse but focused learning outcomes on the idea of classification.

One example of a *Learning* and *Perception* classification game that students developed is *Alien Go* (Fig.5b), where the team designers used the imaginary scenario of meeting extraterrestrial aliens to teach players how supervised ML algorithms work. Team 3 demonstrated their understanding of decision trees through their game description.

Team 3: "So this is our game, it's called *Alien Go*. It's gonna ask you for access to your [smartphone] camera. You gotta walk around and look around for aliens. When you find an alien, it might be one of these [in the drawing], it's gonna be like, 'does it have two eyes? Are they black? So you gotta figure out the classification of the alien... [You win the game] by classifying all the aliens... If you see a new alien and it don't look like nothing you've seen before, you gonna enter [a new alien] in... So you gotta be like, it has legs. It doesn't have legs. You have to input everything about the alien so that it can classify it."

Another example of a game that teaches these topics is *Flower Power* (Fig.5a), where the team designers used the theme of real-world plant classification to teach the potential applications of *Perception* and *Learning* topics for crickets. Team 2 demonstrate their understanding of how machine learning and training models work.

Team 2: "So our game's name is *Flower Power*. And sort of the thought process was that you're basically a cricket. And so you're hopping around like a little garden or a forest or basically anywhere which has plants in it. And based upon that, you're sort of like the machine and you're taking in the information. So every time you see a plant, you would sort of like note the features down..."

These games demonstrate correctly modeled supervised ML classification along with appropriate use cases for them, and draw upon many different themes to expand upon each game to make them more engaging to their peers.

Similarly, during the *Representation & Reasoning* module, students developed mostly single-player roleplaying games involving exploration of a given or generated map. Teams deliberately incorporated topics we taught them, such as Breath-First Search and obstacle avoidance, to explore a city and avoid obstacles. Many of these exploration games involved elements to motivate or engage players, such as survival against threats, or horror elements like game features that scare or unsettle players.

An example that shows how students utilized creative elements in the *Representation & Reasoning* module emerged in Team 5's *Global Guppies* (Fig.1) which takes place in an open ocean to teach *Representation & Reasoning*. Players roleplay as an AI-powered anglerfish in an ocean with different lighting conditions, and are tasked with escaping an orca and avoiding ocean litter by using sensors like LIDAR and camera. Players are later guided by a non-playable character (NPC) dolphin which uses GPS underwater to help them find their way home.

In the *NLP* module, students generally created competitive puzzle-solving board games to teach AI speech detection and NLP. Students'

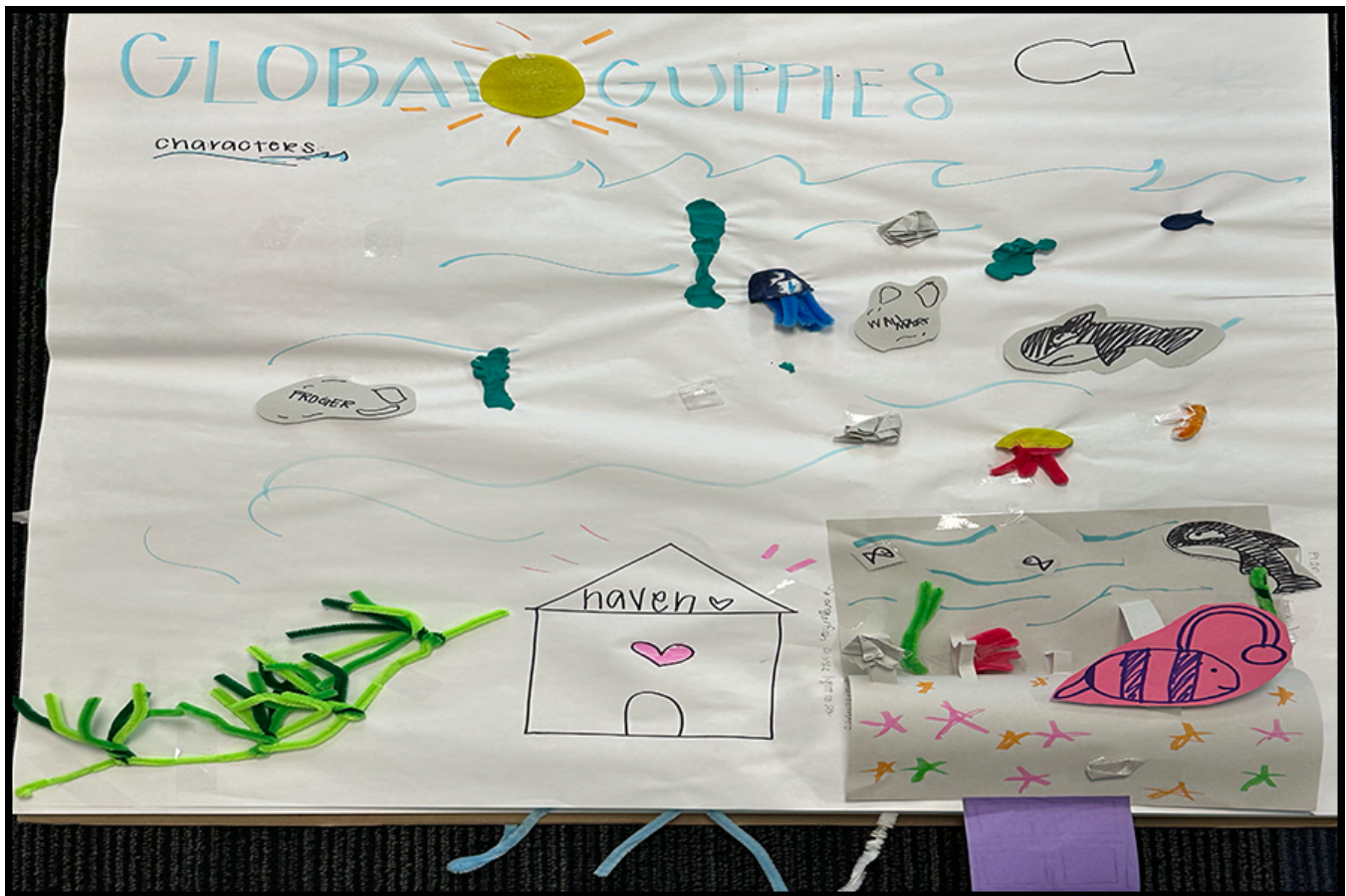


Figure 1: *Global Guppies*, created by Team 4 to teach *Representation & Reasoning*. Players roleplay as an AI-powered anglerfish using LIDAR and camera sensors to escape from a dolphin chasing the player.

games demonstrated understanding for different correct use cases for NLP and AI speech recognition, but were particularly focused on training AIs to parse semantic nuances in ambiguous words or phrases.

For example, *Square Up* is a multiplayer board game where the player draws a card which has a word and its associated context, and the player acts as a classifier to identify if the word is semantically positive or negative. Students deliberately incorporated slang words, such as “fire”, which either means “impressive” or “danger” depending on its usage context, as an example where the AI classifier can be incorrect but the human classifier can be correct. Players are rewarded with in-game currency for correct classification, and the player with the most currency at the end of the game wins.

Across these modules, students’ negotiation processes and final game artifacts illustrate how students ideate different real or imaginary scenarios, and asked players to roleplay as different agents within a broader game narrative to teach AI-related technologies.

4.1.2 Co-designing AI games exposes areas where students may perpetuate underlying AI misunderstandings or biases. We observed instances where students created game narratives or utilized mechanics that demonstrated underlying AI misunderstandings or

had insufficient content knowledge, sometimes by over-simplifying educational game details or neglecting other possible perspectives.

In the game *Global Guppies* (Fig.1), students ask players to roleplay as an anglerfish in a dark ocean using LIDAR and GPS to avoid threats or ocean trash. This game shows students’ mixed levels of understanding of how different technologies can be utilized in different settings, since LIDAR has only limited use underwater [107], and GPS is not at all effective [89]. Such insights provide critical knowledge for instructors to clarify misunderstandings about the limitations of different sensors, in ways that students may not be able to communicate without the process of co-design or creating educational games.

Another example of AI knowledge misconceptions can be seen in the ethics game, *Match the Correct Object with the Action* (Fig.2a). This game can be interpreted as a naive understanding of *AI Ethics* because it assumes all players can agree on what action is “objectively” best for society. The competitive component to this game may also be inappropriate, as it may lead to players competing over one ethical outcome. This may lead to a misunderstanding that there is correct action and object pair to address ethical issues AI

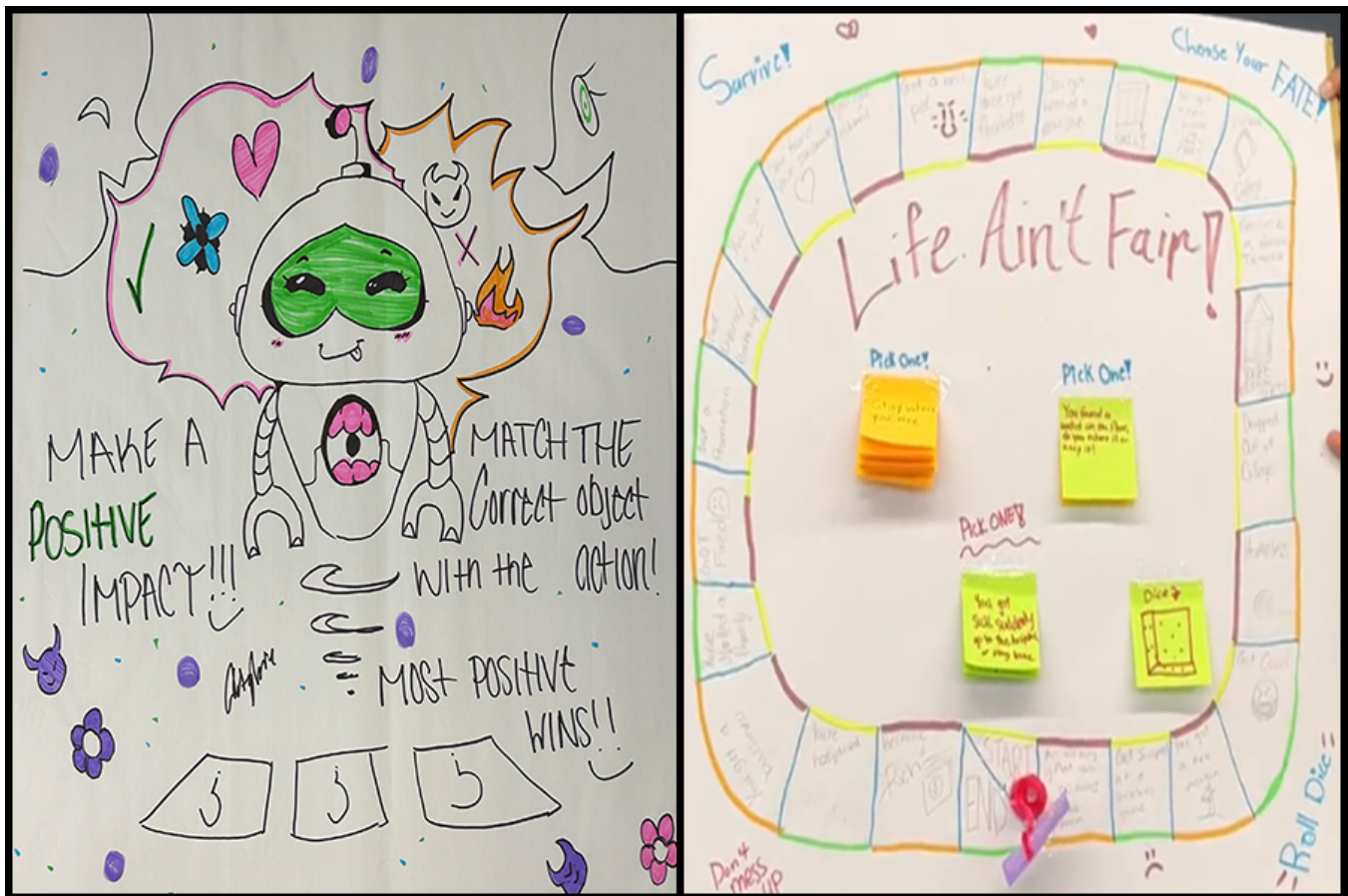


Figure 2: Left: *Match the Correct Object with the Action*, a strategy card game to show how robots can learn to make positive societal impacts using Machine Learning, designed by Team 3. Right: *Life Ain't Fair?*, a board game that asks players to reflect on the ethical dimensions of AI agents making decisions in people's everyday lives. Both of these games were developed to teach AI Ethics.

agents pose in real-world scenarios rather than the the understanding that all actions, taken by AI or humans, bear different levels of risk.

Finally, students may not fully understand the importance of critical elements of game narratives to avoid the perpetuation of bias. One example is *Where's My People*, which is a single-player puzzle-solving roleplaying game designed to teach *Perception* and *Computer Vision*. Players are asked to roleplay as a child who wakes up with amnesia and wants to celebrate her birthday but does not remember who her family is. The child partners with an AI-powered robot sidekick who has clues about who her family is, including a picture of them. It uses a camera as a sensor to take pictures of people as they navigate an open world, and uses image training and ML to identify people who look like the child's family based on the picture clue. When they successfully identify all members of the family, the child can finally celebrate their birthday and the player wins.

Without the inclusion and emphasis of the detail that a picture clue exists, players may assume that the robot is identifying people

based on superficial features such as skin tone or clothing. Such detail may seem minor to young students but are critical because excluding them can perpetuate biases that people are related simply based on their skin tone or people of certain cultures only wear particular clothing. Such game narratives provide educators with wonderful opportunities to discuss the potential for AI to unintentionally perpetuate biases that are ultimately harmful to already marginalized populations.

4.2 RQ2: What does co-design reveal about students' perceptions of the role of AI in educational games?

Our analyses showed that youth understood that AI could serve different roles in their co-designed games. This was manifested in three main narrative interpretations of AI: AI as a helping tool, a playable protagonist, and an ethical agent.

4.2.1 AI as a Helping Tool. Students interpreted AI as a helping tool to accomplish an in-game objective in many of the co-designed

games. These games utilized AI as an additional tool to help the main player accomplish in-game goals, a non-playable character (NPC), or for some other broader in-game objective. This interpretation of AI was prevalent across a majority of games in all units.

The two games in Fig.3 are examples of how AI can be utilized as a helpful tool for achieving goals for different stakeholders, despite having different game and learning objectives. *Recipe Rush* (Fig.3a) is a single-player game where the player is asked to roleplay as a farmer and train a robot classifier that uses cameras and supervised ML to correctly identify crops to create new recipes. AI is an important helping tool to identify different types of crops and suggest new recipes for the farmer to create. Similarly, *Catatouille* (Fig.3b) is a single-player game where the player is asked to roleplay as an AI agent that supplies directions to the main character, a rat, that needs to find its way through a restaurant's kitchen using RADAR and GPS. Both of these games show how youth conceptualize AI agents as helpful tools for stakeholders to accomplish goals in two different narrative stories and learning outcomes.

4.2.2 AI can be a Playable Protagonist. Other times, youth envisioned the complete embodiment of AI agents (rather than an additional tool) as appropriate ways to teach AI concepts. Teams interpreted AI as being a playable character, that required human players to interchangeably take on the role of how an AI agent would act while also considering their own human knowledge and lived experiences.

This perspective is exemplified in *Global Guppies* (Fig.1, described in RQ1) where the main character is an AI-powered anglerfish trying to navigate a dark ocean, avoid an orca and other ocean obstacles to find its way home. Students demonstrated that they understood that more than one AI agent can collaborate to achieve in-game goals – the main player anglerfish is equipped with LIDAR and yet must collaborate with a GPS-powered dolphin to make it home. This game demonstrates students' creativity in creating a broader narrative of roleplaying as an AI, combining multiple uses and embodiment of AI agents to ensure the anglerfish's survival.

Finally, *Reduce, Reuse, Recycle* (Fig.4) is a single-player game that asks players to take on the role of an AI-powered robot that picks up trash in an urban environment. Players control a robot to pick up litter in an urban city, and classify them as recyclable, compostable, or trash. Players are rewarded with in-game currency commissions for correct classifications, but get fired by their human boss if they don't earn any commission after a set period of time. This game is similar to *Global Guppies* in how it envisions a single-player experience of a protagonist roleplaying as an AI, but features more human responsibility in the AI's behavior while striving for positive environmental change.

4.2.3 AI can be an Ethical Agent. Expanding upon Section 4.1.5's descriptions of ethics games, students demonstrated different levels of understanding of AI ethics. Some teams demonstrated an understanding that AI agents can be malleable or changed to be beneficial for society, while others recognized that AI applications can be potentially harmful and asked players to reflect on different perspectives.

Team 3's *Match the Correct Object with the Action* (Fig.2a) demonstrated students understanding that that AI agents can be made

ethical through additional, more diverse training data. This game shows how students thought that biased AI decisions can be fixable through better ML supervision, and how AI could be an ethical agent in helping create decisions that improve society. In contrast, Team 5's *Life Ain't Fair* features players competing and winning by making quiz-based AI decisions that allow players to survive the longest (Fig.2b). It demonstrates a nuanced perspective in their game by asking players to reflect on the ethical dimensions of AI agents making decisions in people's everyday lives. During the presentation of this game, Team 5 stated:

Team 5: "[The point of the game] is to show you that AI is not all knowing. It can only give you answers based on the things that you give it, and the decisions that you give it help pick what your future would be like."

The game demonstrates how players ensure their self-preservation through self-serving AI-made decisions. The competitive nature of the game allows players to make decisions that benefits them despite possible negative impacts to the society as a whole. Both games show nuance in youth perspectives in recognizing that AI can make decisions that have positive and negative societal impacts, while recognizing that AI is not impartial or morally just.

4.3 RQ3: What prior knowledge do students draw upon for AI-related educational game design?

Despite all teams having multiple students with prior CS experience, they did not appear to use this prior CS knowledge when co-designing games. We observed no co-designed games that leveraged explicit CS-related topics (e.g. programming as an input modality) that weren't taught in our sessions; instead, students heavily leveraged their conceptual knowledge of CS such as understanding that robots are programmable and can make decisions based on human input. Students also leveraged their *prior game experience* and *cultural knowledge* to design their own educational AI games.

4.3.1 Leveraging Prior Game Experience. Students leveraged their prior experiences playing video, card, board, or other types of games when designing their own games to teach AI. While some computer games created during the *Perception, Representation and Reasoning*, and *Learning* modules utilized unique game narratives, the board games designed for NLP and Ethics were more directly inspired by board games that students played before.

For example, the games in Fig.5 display how students use prior game experiences to inform their game design and gameplay mechanics to teach AI. *Flower Power* (Fig.5a) made by Team 2 requires the player to correctly classify flowers based on their traits and grow their flower garden to teach ML. The team described its gameplay as a farming simulator; this mechanic was said to be inspired by the publicly available simulation game called *HayDay*.¹ In the same session, Team 3 developed *Alien Go* (Fig.5b). After deliberating on gameplay mechanics, the team leveraged both the artistic

¹<https://hayday.com/en>

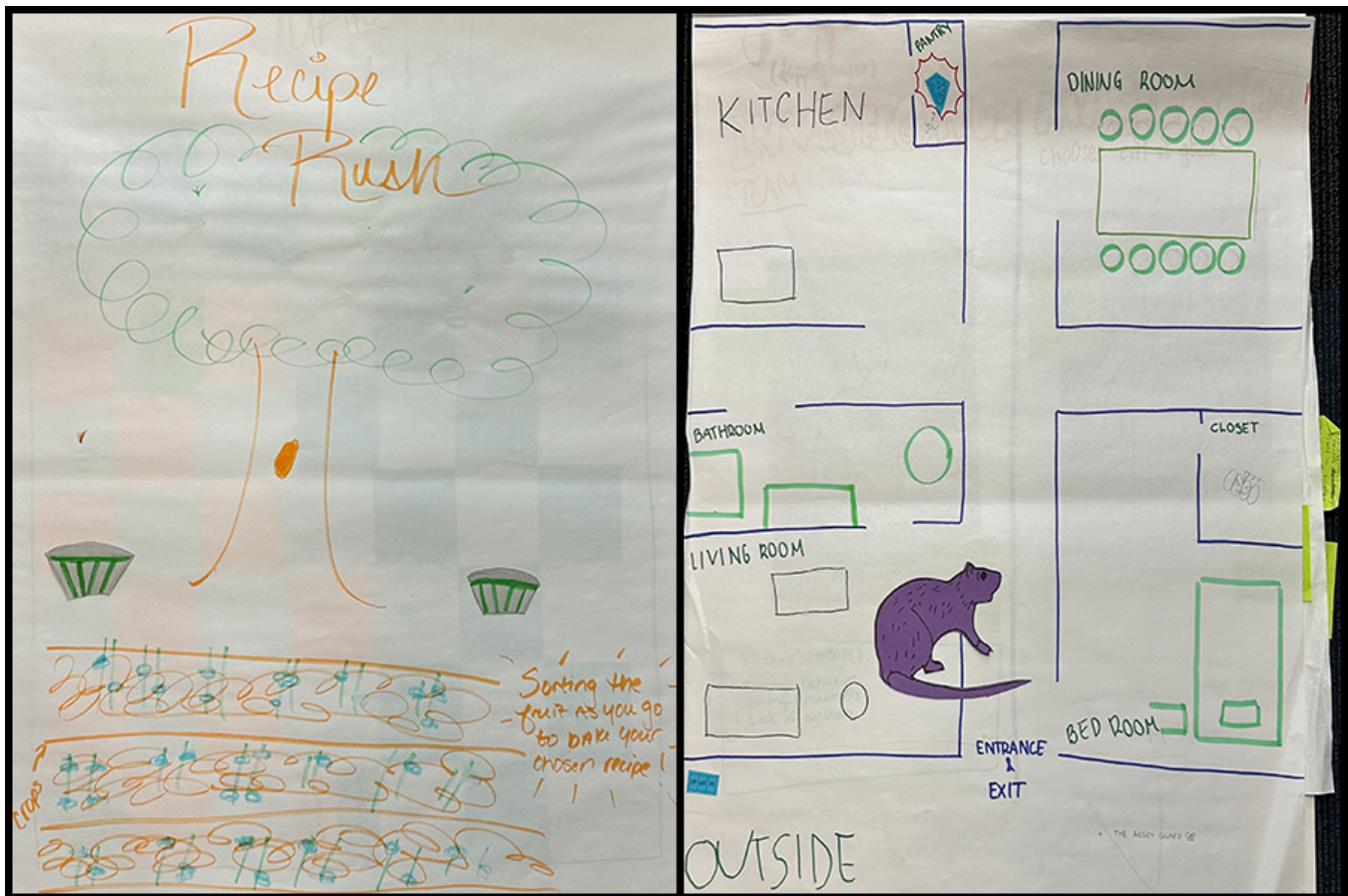


Figure 3: Left: *Recipe Rush*, a single-player game that teaches how farmers can train robots to use image classifiers to identify different crops, designed by Team 1 to teach *Representation & Reasoning*. Right: *Catatouille*, a single-player game that asks players to roleplay as an AI agent to help guide a rat into a restaurant’s kitchen, designed by Team 7 to teach *Perception*.

concepts of the Japanese manga² series *One Piece*³ and the game-play mechanics of the mobile game *Pokémon Go*,⁴ demonstrating how teams drew from multiple games to teach the same topic.

These games also show how students drew from different forms of media references to design AI educational games, taking inspiration from video games series in combination with other media such as novels and manga.

4.3.2 Leveraging Cultural Knowledge. Students drew from their understanding of popular cultural knowledge, such as movies, books, memes, or slang to express their own creativity in imagining unique ways to teach AI-related topics to others.

During Team 1’s creation of *Square Up*, a multiplayer, NLP-based puzzle-solving game, they deliberately chose to use slang as an example of training and de-biasing NLP systems. For example, they said during deliberation:

Team 1: “What’s some good words that are misunderstood easily? Wait, what if we do like slang words? [...]

²Manga refers to Japanese-style graphic novels and comic books.

³https://en.wikipedia.org/wiki/One_Piece

⁴<https://pokemongolive.com/?hl=en>

Like] this sentence, ‘That is so fire’... Ohh how about ‘ate’? Like [in the sentence], ‘you ate that up’... or ‘slay’.”

In this context, the team used slang words. “Ate” is a slang word for conveying approval towards someone or something, or also means that they literally ate something. Similarly, “slay” means performing exceptionally well or impressing someone, or also means to kill something. The co-creation process provides evidence that students of this team drew from their own cultural and linguistic knowledge to enrich the overall gameplay and educational outcomes.

Other times, teams used cultural references to books and other media for inspiration. For example, Team 3 designed *Escape or D13* (also pronounced “Die”), where they deliberated using multiple cultural references: *Five Nights at Freddy’s* (a horror video game series),⁵ *Skibidi Toilet* (an online YouTube series and popular Internet meme),⁶ HP Lovecraft’s work *The Call of Cthulhu* (an American horror short story),⁷ or various monster characters in *Terraria* (an

⁵<https://fivenightsatfreddys3.com>

⁶https://en.wikipedia.org/wiki/Skibidi_Toilet

⁷https://en.wikipedia.org/wiki/The_Call_of_Cthulhu

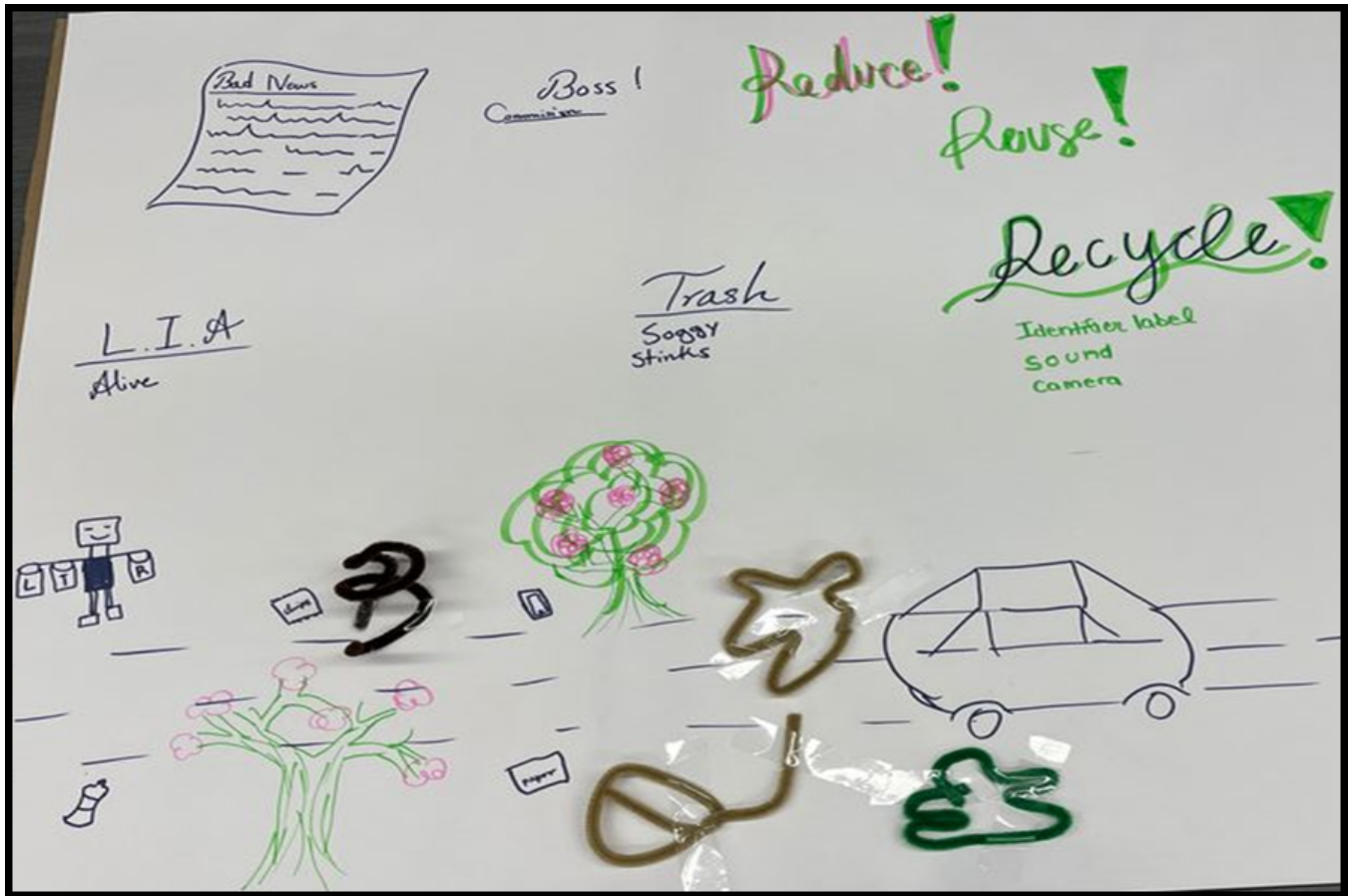


Figure 4: *Reduce, Reuse, Recycle*, a single-player game developed by Team 6 to teach *Learning*. Players train an AI-powered robot to pick up trash and classify trash between recyclable, compostable, and waste.

adventure video game).⁸ The premise of this game is for players to use AI technologies to help them navigate and escape a burning building, where they are trapped with the cosmic entity *Cthulhu* (pronounced Kuh-thool-hu)⁹. The title of the game, *Escape or D13*, also draws inspiration from dice rolls used in the tabletop game *Dungeons & Dragons*¹⁰.

These examples show how students across many different teams combined cultural references, memes, metaphors, language, and concepts as ways to teach AI.

5 Discussion

Our study highlights the benefits of co-designing games to uncover how much students can understand and apply fundamental AI concepts (RQ1). Students went beyond what we taught them, constructed compelling game narratives and used storytelling to envision many different ways AI technologies can be used in real-world applications, such as slang disambiguation, way-finding in different contexts, or de-biasing AI for positive societal impacts.

However, youth sometimes had misunderstandings of applicability of AI concepts, which provide critical insight for AI topics that require additional instructional scaffolding. In terms of how the process of co-design revealed youth participants' emergent perspectives of AI (RQ2), we identified that youth understood how AI can serve different roles in their co-designed games. Youth believed AI could be used as a helpful tool to accomplish in-game objectives, created games that taught AI by embodying AI agents, and also believed that AI agents could have human-like qualities. Lastly, youth believed that AI could be an ethical agent that humans could control or embody to create positive or negative societal impacts. Finally, we identified that youth participants drew from two sources of prior knowledge in constructing AI educational games (RQ3): their *cultural knowledge* and *prior game experiences*. Youth participants combined these sources of knowledge, along with the AI concepts they learned, to create complex, culturally-relevant game elements that enhance the game's enjoyment and contextualize learning outcomes.

In the following sections, we build upon insights from our findings to discuss youth's unique perspectives in AI education, on AI's

⁸<https://terraria.org>

⁹<https://en.wikipedia.org/wiki/Cthulhu>

¹⁰<https://www.dndbeyond.com/how-to-play-dnd>

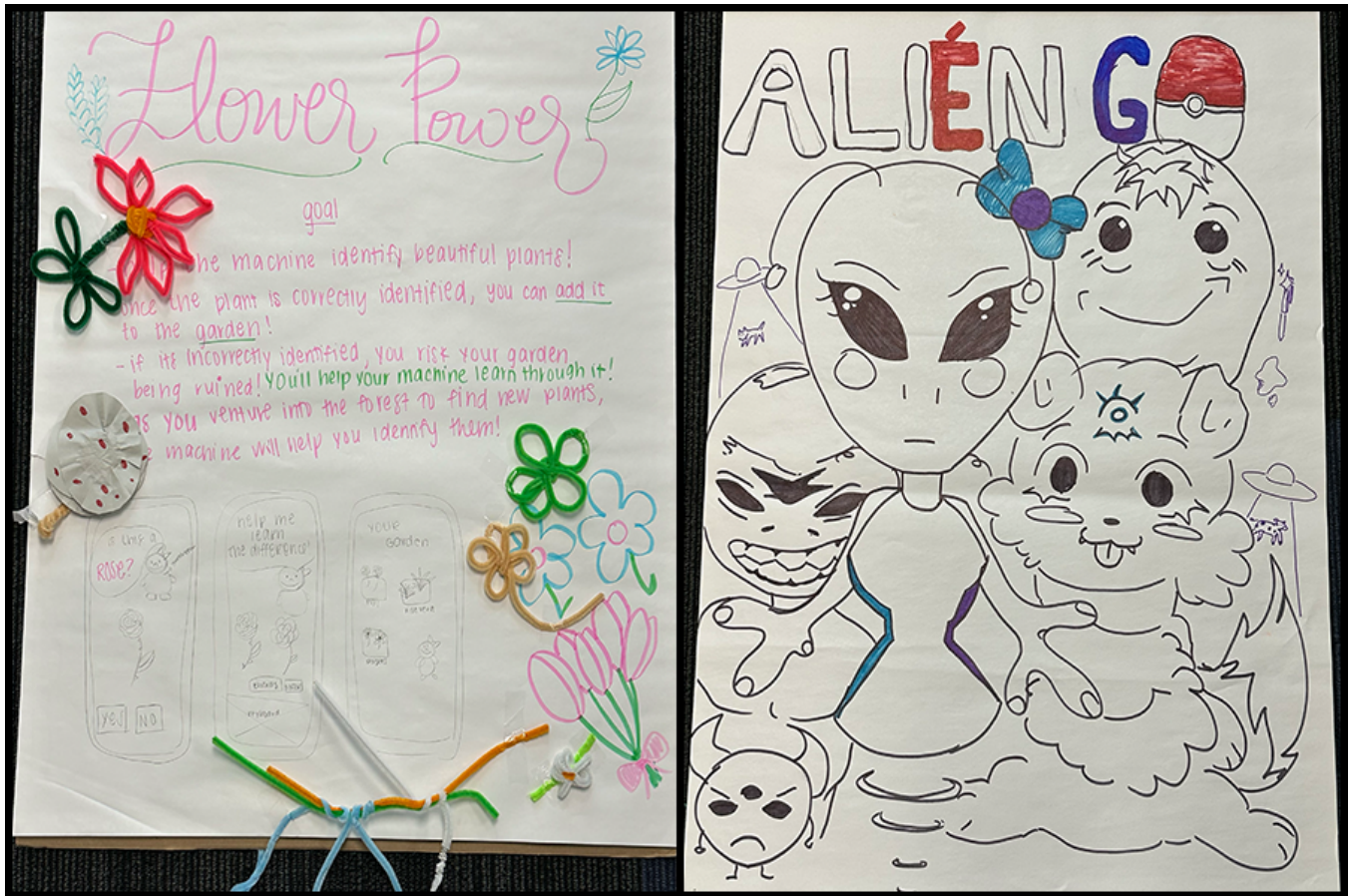


Figure 5: Left: *Flower Power*, a single-player simulation game where players train a classifier to classify flowers in their garden based on flowers' traits. Right: *Alién Go*, a single-player game to teach how players can use photos and image classifiers to classify different alien species apart. Both of these games teach *Learning* and *Perception*.

impact on society, and how AI education could be used as a vehicle for more inclusive CS education.

5.1 Co-designing Educational Activities as a Vehicle for Uncovering Youth's Unique Perspectives of AI in AI Education

Co-designing educational activities proved effective in uncovering youth's unique perspectives in AI education, which reifies prior work that supports co-design to empower youth voices in AI education [61]. Through co-design, our participants combined cultural knowledge and prior game experiences to create rich storytelling narratives. Prior work suggests that these elements are not common in AI or CS educational game design.

Researchers have synthesized literature reviews on the prevalence of game genres in AI and CS education, and have identified RPGs (role-playing games) and competitive games as comparatively uncommon game elements to teach AI and CS topics [94, 106]. Zhan et al. postulate that game elements, such as narrative, fantasy, or competition elements, may create distractions from learning and it is difficult to align game elements with learning content [106].

Contrary to these findings, our youth participants found game elements, particularly narrative and competition, to be helpful ways to contextualize AI learning outcomes. Our findings suggest that including youth in the design of educational games could uncover creative ways for educational game designers to incorporate engaging game elements, such as games featuring narrative and competition, in GBL and AI contexts. We suggest that researchers further investigate youth-led game element design choices as a way to enhance potential learning outcomes and engagement in educational games.

In addition to building narratives, youth conceptualized AI agents as helpful tools, playable protagonists, and ethical agents. The use of AI as a helpful tool is common in many GBL AI educational games. For example, in *AI4Oceans*, players are given a prompt to train an AI to identify fish or trash [16]. However, utilizing AI as a playable protagonist such as in *Global Guppies* or *Reduce, Reuse, Recycle* (see Sections 4.1.2 and 4.2.2 for game descriptions), is a perspective that is uncommon in popular AI education games or tools. This is supported by prior work, where Belghith found that the embodiment of AI agents by youth can help provide enhanced learning opportunities in AI education [8]. *Designing educational*

games where players embody AI agents provides research opportunities to study future learning scenarios of how the roleplaying or embodiment of AI agents can enhance student understanding of AI interactions in the world.

Co-designing educational games with the youth in our study uncovered underlying misconceptions about AI among students. This finding supports prior research that supports co-design's ability to reveal misconceptions in different educational contexts [32, 36]. We show that these insights also apply for evaluating students' AI literacy. Prior work has often used formative assessment in traditional classrooms, or qualitative methods such as qualitative surveys and interviews with students [8, 9, 38, 51, 63, 64, 98] to uncover misconceptions. In CS education contexts, Veerasamy et al. evaluated students' programming misconceptions through formally evaluating students' classwork [98]; In the AI education context, Kanoh used formative assessments by evaluating students' projects and their exploratory behaviors while training machine learning models [51]. A significant amount of prior work also focuses on using semi-structured or focus group interviews to uncover these misunderstandings [8, 63, 64]. Our results suggest that co-design can provide an alternative or complementary method to uncover students' misconceptions without relying on formal assessment or interviews. Co-design can help students express their understanding creatively and collaboratively, which could potentially help reveal misconceptions more authentically. *Our findings support co-design as a valuable way to uncover students' misconceptions, and we suggest that future work can look at using co-design to uncover students' AI misconceptions where formative assessments or interviews are impractical.*

5.2 Youth's Perspectives on AI Bias and AI's Impact in Society

Despite the open-ended nature of our design challenges, students were excited to create educational games that teach the ethical implications of AI technologies. In particular, students enjoyed creating competitive games that rewarded players for maximizing societal benefits and minimizing societal harms. This finding is supported by prior research by Solyst et al. that show that involving youth in co-design workshops to discuss bias in AI allows them to articulate their perspectives and relate algorithmic bias to their lived experiences [85]. Competitive multiplayer games with broader ethical scenarios require players to sometimes attribute human qualities to AI agents, which can be useful for stimulating players to reflect on the societal impacts of AI, and forces players to discuss the scenarios that AI could be beneficial for society. *However, the competitive nature of some games construe "winning" as creating the most "correct" or societally utilitarian classifiers, which can be a naive interpretation that there is a clear "correct" approach to societal issues and that it is possible for AI to make these decisions.* In addition, competitive games could lead to students prioritizing one perspective over another to maximize their chances of winning. *This implies that collaborative games where all players work together for the common good may be more effective for helping students focus on more fair uses of AI for societal good.*

Youth's eager engagement with AI bias and ethics related topics in our study supports previous calls for embedding AI ethics

education into AI literacy programs [58], especially when they are given the space and opportunity to do so [17, 56, 85]. Coenraad demonstrated how youth are visibly aware of the negative impacts of computing technologies, but did not have a formal vocabulary to discuss technological biases [17]. Other work by Landesman & Salac et al. shed light on how scaffolded instruction around technological sensemaking can lead to youth being capable of attaining formal vocabulary in identifying fair and unfair treatment from AI [56]. While prior work supports instructor-led scaffolding of different ethical perspectives students can use, our participants imagined how educational games could also be used to adopt different ethical perspectives through the embodiment of AI agents. *We suggest that multiplayer GBL activities could be made more effective by explicitly prompting players to consider their own ethical perspectives in the design of educational games. In addition, future research can explore how to scaffold multiplayer games as launching pads for ethical reasoning around AI, intentionally designing them to be sufficiently nuanced to elicit deeper ethical sensemaking in students or players.*

5.3 Incorporating Diverse Youth Voices into AI Literacy Education

Incorporating students' voices into the negotiation and design of educational game elements can help designers incorporate culturally relevant design choices for diverse student audiences. Adult-designed educational games have been used in AI education [106]; however, they often rely on re-purposing entertainment-purpose games to maintain engagement for younger audiences [67, 106]. Prior work has controversially tried "girl games" such as *Barbie Fashion Designer*¹¹ or incorporating stereotypically feminine features into games like *The Sims 2* [44] to appeal to girls. Such initiatives may reinforce gender stereotypes in educational games [22, 67], and impede efforts to improve STEM engagement for Black women and girls by reinforcing harmful stereotypes [30, 48, 74, 75, 84, 90]. As a response, researchers such as Joseph and Thomas [48, 90] have actively involved Black youth in game design initiatives. Such efforts have successfully positioned Black girls as producers of CS knowledge, counter-narrative to historical perceptions of deficit-model intellectual gaps in Black girls' ability to learn CS [48, 90]. In AI education, Rankin & Henderson show that involving Black youth in the ideation process of re-imagining conversational agents led to design visions for conversational agents that better reflect their own racial and gender identities [74]. In co-design workshops about AI fairness and ethics, Black girls are able to recognize and critique systems of oppression and understand how AI can impact their communities, such as expressing concern for how algorithms could help support mixing Black and White students into the same schools, rather than segregating them [84, 86].

Our work reifies such research efforts by demonstrating how involving youth in the design of educational activities lends agency and builds counter-narratives to predominant deficit-based stereotypes, and offers appropriate design insights to increase engagement for girls of color. None of the youth-led co-designed games display characteristics of "girl game" elements or prescriptively feminine characteristics [22, 67]. Instead, they feature how youth

¹¹https://en.wikipedia.org/wiki/Barbie_Fashion_Designer

incorporate diverse cultural knowledge into the design of educational AI games. *Future work should further involve students of color in designing games that appeal to diverse audiences and reflect asset-based ways of empowering them.* Future work can also consider employing critical lenses, such as *intersectionality* [73] to understand better how to empower underrepresented youth's experiences in game design.

5.4 AI Literacy as a Vehicle for CS Education

Our insights also suggest that youth students do not need explicit CS or programming-intensive prior education to develop their AI literacy, making AI education more accessible to students who do not have strong CS backgrounds. Our co-design teams did not explicitly utilize their prior CS knowledge to teach AI concepts and did not expect players to have any background in either CS or AI concepts. This is not to say CS education is not beneficial to AI literacy – students utilized computational thinking skills in their game design choices e.g., in *Where's My People* where the robot helps the protagonist solve different clues and classifying subsets of the picture clue to ultimately find the child's family. However, these insights can be obtained from other subject areas and not just in computer science, alleviating the need for CS courses to be a prerequisite for students engaging in AI education. This insight is in contrast with prior literature reviews on AI education that find that many AI education curricula in both K-12 and tertiary levels focus heavily on technical programming or make it a requirement before learning AI [57, 102, 106]. Such studies include Zhan et al.'s findings on AI education interventions that use GBL, where they found that these interventions focused heavily on computer programming while basic AI conceptual knowledge was comparatively neglected [106]. *This suggests that students can improve their understanding of the real-world applications of AI technologies by engaging in activity- and critical reflection-based instruction without relying on their prior CS knowledge as a prerequisite.*

Finally, our research findings suggest that unplugged, low-tech educational activities can be beneficial for increasing students' exposure to AI concepts and further STEM and CS education in low resource and informal settings. Notably, our co-design sessions were non-technical, low-resource, and focused on students' own creativity, *offering opportunities for educators seeking to expand AI literacy to low-resource communities.* Prior work has explored using informal learning environments, such as after-school care centers, as places to teach STEM topics to low-resource youth [59, 92]. Scholars have found that such settings can be beneficial for youth's development of STEM identity, or sense of belonging in the field, and are beneficial for students' interest in pursuing future STEM careers [5, 13]. Our work reifies prior work in supporting how non-traditional classrooms can support students' learning and potential interest in the field. Our study settings allowed for students' creative expression and close engagement in the GBL design activities. The co-design activities highlight how AI applications, such as LIDAR, were engaging enough for youth of diverse cultural backgrounds and mixed levels of CS knowledge to create fun activities to teach AI concepts and applications. *From these insights, we argue that AI literacy can be utilized in low-resource settings as a vehicle to introduce students to CS or STEM in low-resource or non-traditional*

learning environments. Future work can explore how youth-created GBL can support students' interest in AI and improve their STEM identity and sense of belonging.

5.5 Limitations and Future Work

In our study, co-designing educational games provided creative outlets for youth to express alternative ways to imagine AI education. However, prior critical work on Participatory Design by Harington & Erete note that PD itself is a privileged activity, and that utilizing this research method may perpetuate inequity, given that PD can situate researchers as holding a disproportional amount of design and technological knowledge [41]. We encourage researchers to critically consider the assets that diverse participants bring to co-design settings to ensure that all participants contribute equitably to the design process and output. We also recognize that our youth participants were enrolled in a STEM-internship program on a STEM-focused academic institution; they likely already developed STEM identities, and do not represent the general youth population. Finally, we recognize that we did not engage with youth in their own situated spaces. Engaging in this research effort in environments where youth have to contend with other responsibilities from their families, schools, or communities may lead to different research insights.

Our team is actively developing AI educational games based on the presented co-designed game ideas and are playtesting them with youth in similar age groups. Future studies are necessary to determine if these games are effective for improving students' AI literacy in formal and informal learning settings. Finally, future work should continue to explore embedding non-programming AI concepts into K-12 curricula as a vehicle for introducing students to CS and developing their STEM identities. Such activities are simple enough to engage students and pique their interest in AI before more technical topics may be introduced.

6 Conclusion

We facilitated co-design sessions to create educational games that teach AI to thirty-nine Black, Asian, and Hispanic girls and non-binary youth. Our goal was to understand what the co-designed games revealed about students' ability to understand and apply fundamental AI concepts in game-based activities, youths' unique perspectives of AI agents, and the prior knowledge they draw from. We found that youth participants went beyond what we taught them and used narratives to contextualize how AI technologies could be used for real-world applications. Youth imagined players of their games as being able to embody AI agents as a valid way to learn about the capabilities and applications of AI technologies, and also imagined that AI can be an ethical agent to make decisions that impact society. Youth leveraged their prior game experiences and cultural knowledge in designing these educational games. We reflect on the benefits of incorporating youths' perspectives into both game design and AI curricula, and provide suggestions for AI curriculum designers, instructors teaching AI, and game designers to create educational games that are engaging to diverse youth populations.

Acknowledgments

We thank all the participants, teaching assistants, and collaborators in the Georgia Tech BridgeUp program for their partnership to complete this study. Special thanks to Will Gelder, Briana Marks, Elizabeth Ayala Mojica, Druti Naik, and Elliot Roe for their amazing and hard work as co-design facilitators and teaching assistants, and administrator Cedric Stallworth for running this program. We extend our thanks to all who helped read and edit the manuscript.

References

- [1] AI For K-12. 2024. *Artificial Intelligence For K-12*. Artificial Intelligence for K-12. <https://ai4k12.org>
- [2] AI4GA. 2024. *Artificial Intelligence For Georgia*. Artificial Intelligence for Georgia. <https://ai4ga.org/>
- [3] Safinah Ali, Daniella DiPaola, and Cynthia Breazeal. 2021. What are GANs?: introducing generative adversarial networks to middle school students. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35, 15472–15479.
- [4] Safinah Ali, Blakeley H Payne, Randi Williams, Hae Won Park, and Cynthia Breazeal. 2019. Constructionism, ethics, and creativity: Developing primary and middle school artificial intelligence education. In *International workshop on education in artificial intelligence k-12 (eduai'19)*, Vol. 2. MIT Media Lab Palo Alto, California, 1–4.
- [5] Patricia J Allen, Rong Chang, Britt K Gorrall, Luke Waggenspack, Eriko Fukuda, Todd D Little, and Gil G Noam. 2019. From quality to outcomes: A national study of afterschool STEM programming. *International Journal of STEM Education* 6 (2019), 1–21.
- [6] Diego Alvarado. 2012. Supporting non-formal learning through co-design of social games with children. In *Proceedings of the 11th International Conference on Interaction Design and Children (Bremen, Germany) (IDC '12)*. Association for Computing Machinery, New York, NY, USA, 347–350. doi:10.1145/2307096.2307164
- [7] William F Arsenio, Susanna Preziosi, Erica Silberstein, and Benjamin Hamburger. 2012. Adolescents' perceptions of institutional fairness: Relations with moral reasoning, emotions, and behavior. *New directions for youth development* 2012, 136 (2012), 95–110.
- [8] Yasmine Belghith, Atefeh Mahdavi Goloujeh, Brian Magerko, Duri Long, Tom Mcklin, and Jessica Roberts. 2024. Testing, Socializing, Exploring: Characterizing Middle Schoolers' Approaches to and Conceptions of ChatGPT. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 276, 17 pages. doi:10.1145/3613904.3642332
- [9] Arne Bewersdorff, Xiaoming Zhai, Jessica Roberts, and Claudia Nerdel. 2023. Myths, mis- and preconceptions of artificial intelligence: A review of the literature. *Computers and Education: Artificial Intelligence* 4 (2023), 100143.
- [10] Ashly Burch and Rosalind Wiseman. 2015. Curiosity, courage and camouflage: Revealing the gaming habits of teen girls. In *Game Developers Conference*, Vol. 2, 2016.
- [11] Eva Cerezo, Carina S González-González, and Clara Bonillo. 2023. Empowering soft skills in children with ADHD through the co-creation of tangible tabletop games. *Universal Access in the Information Society* 23, 1 (2023), 3–21.
- [12] Yuki Chen, Jonaya Kemper, Erik Harpstead, Ross Higashi, and Judith Uchidiuno. 2023. Designing black children in video games. *Interactions* 30, 5 (2023), 54–56.
- [13] Jessica R Chittum, Brett D Jones, Sehmuz Akalin, and Ásta B Schram. 2017. The effects of an afterschool STEM program on students' motivation and engagement. *International journal of STEM education* 4 (2017), 1–16.
- [14] Thomas KF Chiu. 2021. A holistic approach to the design of artificial intelligence (AI) education for K-12 schools. *TechTrends* 65, 5 (2021), 796–807.
- [15] Diane Coddington, Bataul Alkhateeb, Chrystalla Mouza, and Lori Pollock. 2021. From professional development to pedagogy: Examining how Computer Science teachers conceptualize and apply culturally responsive pedagogy. *Journal of Technology and Teacher Education* 29, 4 (2021), 497–532.
- [16] Code.org. 2024. *AI for Oceans*. <https://studio.code.org/s/oceans>
- [17] Merijke Coenraad. 2022. "That's what techquity is": youth perceptions of technological and algorithmic bias. *Information and Learning Sciences* 123, 7/8 (2022), 500–525.
- [18] Merijke Coenraad, Jen Palmer, Diana Franklin, and David Weintrop. 2019. Utilizing Participatory Design to Develop a Culturally Relevant Computer Science Curriculum. In *Proceedings of the 50th ACM technical symposium on computer science education*, 1261–1261.
- [19] Amanda C Cote. 2020. Gaming sexism: Gender and identity in the era of casual video games. In *Gaming Sexism*. New York University Press.
- [20] Yun Dai. 2023. Negotiation of epistemological understandings and teaching practices between primary teachers and scientists about artificial intelligence in professional development. *Research in Science Education* 53, 3 (2023), 577–591.
- [21] Hojjat Dehghanzadeh, Mohammadreza Farrokhnia, Hossein Dehghanzadeh, Kiumars Taghipour, and Omid Noroozi. 2023. Using gamification to support learning in K-12 education: A systematic literature review. *British Journal of Educational Technology* 55, 1 (2023), 34–70.
- [22] Michele D Dickey. 2006. Girl gamers: The controversy of girl games and the relevance of female-oriented game design for instructional design. *British journal of educational technology* 37, 5 (2006), 785–793.
- [23] Stefania Druga and Amy J Ko. 2021. How do children's perceptions of machine intelligence change when training and coding smart programs?. In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference (Athens, Greece) (IDC '21)*. Association for Computing Machinery, New York, NY, USA, 49–61. doi:10.1145/3459990.3460712
- [24] Stefania Druga, Sarah T. Vu, Eesh Likhith, and Tammy Qiu. 2019. Inclusive AI literacy for kids around the world. In *Proceedings of FabLearn 2019* (New York, NY, USA) (FL2019). Association for Computing Machinery, New York, NY, USA, 104–111. doi:10.1145/3311890.3311904
- [25] Stefania Druga, Randi Williams, Cynthia Breazeal, and Mitchel Resnick. 2017. "Hey Google is it OK if I eat you?": Initial Explorations in Child-Agent Interaction. In *Proceedings of the 2017 Conference on Interaction Design and Children* (Stanford, California, USA) (IDC '17). Association for Computing Machinery, New York, NY, USA, 595–600. doi:10.1145/3078072.3084330
- [26] Stefania Druga, Jason Yip, Michael Preston, and Devin Dillon. 2023. The 4 As: Ask, Adapt. *Algorithmic Rights and Protections for Children* (2023), 193.
- [27] Allison Druin. 1999. Cooperative inquiry: developing new technologies for children with children. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*, 592–599.
- [28] Allison Druin and Jon Kolko. 2017. Conversation: Participatory Design in research and practice. In *Participatory Design for Learning*. Routledge, 189–201.
- [29] ErasmusPlus. 2024. *Learn to Machine Learn*. <https://learnml.eu/>
- [30] Sheena Erete, Yolanda A Rankin, and Jakita O Thomas. 2021. I can't breathe: Reflections from Black women in CSCW and HCI. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW3 (2021), 1–23.
- [31] Deniz Eseryel, Victor Law, Dirk Ifenthaler, Xun Ge, and Raymond Miller. 2014. An investigation of the interrelationships between motivation, engagement, and complex problem solving in game-based learning. *Journal of Educational Technology & Society* 17, 1 (2014), 42–53.
- [32] Caitlin GM Fine and Erin M Furtak. 2024. Asset and deficit discourses of student ideas in science formative assessment co-design. *Journal of Research in Science Teaching* (2024).
- [33] Jesse Fox and Wai Yen Tang. 2014. Sexism in online video games: The role of conformity to masculine norms and social dominance orientation. *Computers in human behavior* 33 (2014), 314–320.
- [34] Geneva Gay. 2018. *Culturally responsive teaching: Theory, research, and practice*. teachers college press.
- [35] Clifford Geertz. 2008. Thick description: Toward an interpretive theory of culture. In *The cultural geography reader*. Routledge, 41–51.
- [36] Gian Andrea Giacobone, Alessandro Pollini, Diana Urquiza, and Konstantina Chouta. 2024. Participatory design methods for sustainable interaction design: co-designing digital experiences for sustainability education. *Universal Access in the Information Society* (2024), 1–17.
- [37] Google. 2024. *Teachable Machine*. <https://teachablemachine.withgoogle.com/>
- [38] Shuchi Grover. 2021. Toward A Framework for Formative Assessment of Conceptual Learning in K-12 Computer Science Classrooms. In *Proceedings of the 52nd ACM Technical Symposium on Computer Science Education*. Association for Computing Machinery, New York, NY, USA, 31–37. doi:10.1145/3408877.3432460
- [39] David Hammer and Leema K Berland. 2014. Confusing claims for data: A critique of common practices for presenting qualitative research on learning. *Journal of the Learning Sciences* 23, 1 (2014), 37–46.
- [40] Michael D Hanus and Jesse Fox. 2015. Assessing the effects of gamification in the classroom: A longitudinal study on intrinsic motivation, social comparison, satisfaction, effort, and academic performance. *Computers & education* 80 (2015), 152–161.
- [41] Christina Harrington, Sheena Erete, and Anne Marie Piper. 2019. Deconstructing community-based collaborative design: Towards more equitable participatory design engagements. *Proceedings of the ACM on human-computer interaction* 3, CSCW (2019), 1–25.
- [42] Leshell Hatley, Cynthia E Winston-Proctor, Gina M Paige, and Kevin Clark. 2017. Culture and computational thinking: A pilot study of operationalizing culturally responsive teaching (CRT) in computer science education. In *Culture, Learning, and Technology*. Routledge, 109–126.
- [43] Martti Havukainen, Teemu H Laine, Timo Martikainen, and Erkki Sutinen. 2020. A case study on co-designing digital games with older adults and children: game elements, assets, and challenges. *The Computer Games Journal* 9 (2020), 163–188.
- [44] Elisabeth R Hayes and Elizabeth M King. 2009. Not just a dollhouse: what The Sims2 can teach us about women's IT learning. *On the Horizon* 17, 1 (2009), 60–69.

- [45] Clint Heinze, Janet Haase, and Helen Higgins. 2010. An action research report from a multi-year approach to teaching artificial intelligence at the k-6 level. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 24. 1890–1895.
- [46] Sofia Hussain. 2010. Empowering marginalised children in developing countries through participatory design processes. *CoDesign* 6, 2 (2010), 99–117.
- [47] Mahmood H Hussein, Siew Hock Ow, Monther M Elaish, and Erik O Jensen. 2022. Digital game-based learning in K-12 mathematics education: a systematic literature review. *Education and Information Technologies* 27, 2 (2022), 2859–2891.
- [48] N Joseph and Jakita O Thomas. 2020. Designing STEM learning environments to support middle school black girls' computational algorithmic thinking: A possibility model for disrupting STEM neoliberal projects. In *International Conference of the Learning Sciences*.
- [49] Kanwar Sunreep Jossan, Andrea Gauthier, and Jodie Jenkinson. 2021. Cultural implications in the acceptability of game-based learning. *Computers & Education* 174 (2021), 104305.
- [50] Yasmin Kafai. 2008. Considering gender in digital games: Implications for serious game designs in the learning sciences. *International Society of the Learning Sciences* (2008).
- [51] Hiroko KANO. 2024. Clarifying Misconceptions: Key AI Concepts for Students in Model Creation Classes. *JSSSE Research Report* 39, 1 (2024), 59–64.
- [52] Maria Kasinidou, Styliani Kleanthous, and Jahna Otterbacher. 2024. "AI is a robot that knows many things": Cypriot children's perception of AI. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference* (Delft, Netherlands) (IDC '24). Association for Computing Machinery, New York, NY, USA, 897–901. doi:10.1145/3628516.3659414
- [53] Machine Learning For Kids. 2024. *Machine Learning for Kids*. <https://machinelearningforkids.co.uk/>
- [54] Priya Kumar, Jessica Vitak, Marshini Chetty, Tamara L. Clegg, Jonathan Yang, Brenna McNally, and Elizabeth Bonsignore. 2018. Co-designing on-line privacy-related games and stories with children. In *Proceedings of the 17th ACM Conference on Interaction Design and Children* (Trondheim, Norway) (IDC '18). Association for Computing Machinery, New York, NY, USA, 67–79. doi:10.1145/3202185.3202735
- [55] G Lampropoulos. 2023. Educational benefits of digital game-based learning: K-12 teachers' perspectives and attitudes. *Advances in Mobile Learning Educational Research*, 3 (2), 805–817.
- [56] Rotem Landesman, Jean Salac, Jared Ordoña Lim, and Amy J Ko. 2024. Integrating Philosophy Teaching Perspectives to Foster Adolescents' Ethical Sensemaking of Computing Technologies. In *Proceedings of the 2024 ACM Conference on International Computing Education Research-Volume 1*. 502–516.
- [57] Sang Joon Lee and Kyungbin Kwon. 2024. A systematic review of AI education in K-12 classrooms from 2018 to 2023: Topics, strategies, and learning outcomes. *Computers and Education: Artificial Intelligence* (2024), 100211.
- [58] Victor R Lee, Victoria Delaney, and Parth Sarin. 2022. Eliciting High School Students' Conceptions and Intuitions about Algorithmic Bias. In *Proceedings of the 2022 ACM Conference on International Computing Education Research-Volume 2*. 35–36.
- [59] Jared Lim, Grace Barkhuff, and Judith Uchidiuno. 2024. Increasing Access to CS Instruction in Low-Income Afterschool Settings. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2* (Portland, OR, USA) (SIGCSE 2024). Association for Computing Machinery, New York, NY, USA, 1720–1721. doi:10.1145/3626253.3635559
- [60] Kip Lim and Colleen M Lewis. 2020. Three metrics of success for high school CSforAll initiatives: Demographic patterns from 2003 to 2019 on Advanced Placement Computer Science exams. In *Proceedings of the 51st ACM Technical Symposium on Computer Science Education*. 598–604.
- [61] Duri Long, Takeria Blunt, and Brian Magerko. 2021. Co-Designing AI Literacy Exhibits for Informal Learning Spaces. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 293 (Oct. 2021), 35 pages. doi:10.1145/3476034
- [62] Livia S Marques, Christiane Gresse von Wangenheim, and Jean CR Hauck. 2020. Teaching machine learning in school: A systematic mapping of the state of the art. *Informatics in Education* 19, 2 (2020), 283–321.
- [63] Erik Marx, Clemens Witt, and Thimo Leonhardt. 2024. Identifying Secondary School Students' Misconceptions about Machine Learning: An Interview Study. In *Proceedings of the 19th WiPSCe Conference on Primary and Secondary Computing Education Research*. 1–10.
- [64] Pekka Mertala and Janne Fagerlund. 2024. Finnish 5th and 6th graders' misconceptions about artificial intelligence. *International Journal of Child-Computer Interaction* 39 (2024), 100630.
- [65] Oussama Metatla, Sandra Bardot, Clare Cullen, Marcos Serrano, and Christophe Jouffrais. 2020. Robots for Inclusive Play: Co-designing an Educational Game With Visually Impaired and sighted Children. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376270
- [66] Scott Nicholson. 2014. *A RECIPE for Meaningful Gamification*. Springer International Publishing, Cham, 1–20. doi:10.1007/978-3-319-10208-5_1
- [67] Amanda Ochsner. 2015. Lessons Learned With Girls, Games, and Design. In *Proceedings of the Third Conference on GenderIT* (Philadelphia, PA, USA) (GenderIT '15). Association for Computing Machinery, New York, NY, USA, 24–31. doi:10.1145/2807565.2807709
- [68] Benjamin Paaßen, Thekla Morgenroth, and Michelle Stratemeyer. 2017. What is a true gamer? The male gamer stereotype and the marginalization of women in video game culture. *Sex Roles* 76 (2017), 421–435.
- [69] Samiullah Paracha, Lynne Hall, Kathy Clawson, Nicole Mitsche, and Fatima Jamil. 2019. Co-design with Children: Using Participatory Design for Design Thinking and Social and Emotional Learning. *Open Education Studies* 1, 1 (2019), 267–280. doi:10.1515/edu-2019-0021
- [70] Woongbin Park and Hyuksoo Kwon. 2024. Implementing artificial intelligence education for middle school technology education in Republic of Korea. *International journal of technology and design education* 34, 1 (2024), 109–135.
- [71] Ana Cristina Pires, Ewelina Bakala, Fernando González-Perilli, Gustavo Sansone, Bruno Fleischer, Sebastián Marichal, and Tiago Guerreiro. 2022. Learning maths with a tangible user interface: Lessons learned through participatory design with children with visual impairments and their educators. *International Journal of Child-Computer Interaction* 32 (2022), 100382. doi:10.1016/j.ijcci.2021.100382
- [72] Joseph G Ponterotto. 2006. Brief note on the origins, evolution, and meaning of the qualitative research concept thick description. *The qualitative report* 11, 3 (2006), 538–549.
- [73] Yolanda A Rankin, Sheena Erete, Jakita O Thomas, and Nichole Pinkard. 2024. The Choice is Yours: Intersectional Studies versus Studies of Intersectional Populations in Computing Education Research. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1*. 1098–1104.
- [74] Yolanda A Rankin and Kallayah K Henderson. 2021. Resisting racism in tech design: centering the experiences of black youth. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–32.
- [75] Yolanda A Rankin, Jakita O Thomas, and Sheena Erete. 2021. Black women speak: Examining power, privilege, and identity in CS education. *ACM Transactions on Computing Education (TOCE)* 21, 4 (2021), 1–31.
- [76] Janet C. Read, Daniel Fitton, and Matthew Horton. 2014. Giving ideas an equal chance: inclusion and representation in participatory design with children. In *Proceedings of the 2014 Conference on Interaction Design and Children* (Aarhus, Denmark) (IDC '14). Association for Computing Machinery, New York, NY, USA, 105–114. doi:10.1145/2593968.2593986
- [77] Jeremy Roschelle, James Lester, and Judi Fusco. 2020. AI and the Future of Learning: Expert Panel Report. *Digital Promise* (2020).
- [78] Jean J. Ryoo. 2019. Pedagogy that Supports Computer Science for All. *ACM Trans. Comput. Educ.* 19, 4, Article 36 (jul 2019), 23 pages. doi:10.1145/3322210
- [79] Jean Salac, Alannah Oleson, Lena Armstrong, Audrey Le Meur, and Amy J Ko. 2023. Funds of Knowledge used by Adolescents of Color in Scaffolded Sensemaking around Algorithmic Fairness. In *Proceedings of the 2023 ACM Conference on International Computing Education Research-Volume 1*. 191–205.
- [80] Ismaila Temitayo Sanusi, Solomon Sunday Oyelere, Henriikka Vartiainen, Jarkko Suhonen, and Markku Tuikiainen. 2023. A systematic review of teaching and learning machine learning in K-12 education. *Education and Information Technologies* 28, 5 (2023), 5967–5997.
- [81] Liuyi Shu and Min Liu. 2019. Student engagement in game-based learning: A literature review from 2008 to 2018. *Journal of Educational Multimedia and Hypermedia* 28, 2 (2019), 193–215.
- [82] Zoe Skinner, Stacey Brown, and Greg Walsh. 2020. Children of color's perceptions of fairness in AI: An exploration of equitable and inclusive co-design. In *Extended abstracts of the 2020 CHI conference on human factors in computing systems*. 1–8.
- [83] Jaemarie Solyst, Judith Odili Uchidiuno, Erik Harpstead, Jonaya Kemper, and Ross Higashi. 2023. Comparative Design-Based Research: How Afterschool Programs Impact Learners' Engagement with a Video Game Codesign. *arXiv preprint arXiv:2307.11903* (2023).
- [84] Jaemarie Solyst, Shixian Xie, Ellia Yang, Angela EB Stewart, Motahhare Eslami, Jessica Hammer, and Amy Ogan. 2023. "I Would Like to Design": Black Girls Analyzing and Ideating Fair and Accountable AI. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [85] Jaemarie Solyst, Ellia Yang, Shixian Xie, Amy Ogan, Jessica Hammer, and Motahhare Eslami. 2023. The Potential of Diverse Youth as Stakeholders in Identifying and Mitigating Algorithmic Bias for a Future of Fairer AI. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 364 (Oct. 2023), 27 pages. doi:10.1145/3610213
- [86] Jaemarie Solyst, Ellia Yang, Shixian Xie, Amy Ogan, Jessica Hammer, and Motahhare Eslami. 2023. The Potential of Diverse Youth as Stakeholders in Identifying and Mitigating Algorithmic Bias for a Future of Fairer AI. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 364 (Oct. 2023), 27 pages. doi:10.1145/3610213
- [87] Steven J Spencer, Christine Logel, and Paul G Davies. 2016. Stereotype threat. *Annual review of psychology* 67, 1 (2016), 415–437.
- [88] Anselm Strauss and Juliet Corbin. 1998. Basics of qualitative research techniques. (1998).

- [89] Gunnar Taraldsen, Tor Arne Reinen, and Tone Berg. 2011. The underwater GPS problem. In *OCEANS 2011 IEEE - Spain*. 1–8. doi:10.1109/Oceans-Spain.2011.6003649
- [90] Jakita O Thomas, O Carlette Odemwingie, Quimeka Saunders, and Malika Watlerd. 2015. Understanding the difficulties African-American middle school girls face while enacting computational algorithmic thinking in the context of game design. (2015).
- [91] David Touretzky, Christina Gardner-McCune, Fred Martin, and Deborah See-horn. 2019. Envisioning AI for K-12: What should every child know about AI?. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 33. 9795–9799.
- [92] Judith Odili Uchidiuno, Jaemarie Solyst, Erik Harpstead, and Ross Higashi. 2023. "I'm a little less inclined to do it": How Afterschool Programs' Culture Impact Co-Design Processes and Outcomes. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference (Pittsburgh, PA, USA) (DIS '23)*. Association for Computing Machinery, New York, NY, USA, 1263–1276. doi:10.1145/3563657.3596006
- [93] UNICEF. 2021. *Policy Guidance on AI for Children*. <https://www.unicef.org/globalinsight/media/2356/file/UNICEF-Global-Insight-policy-guidance-AI-children-2.0-2021.pdf>
- [94] Adilson Vahldick, Antonio José Mendes, and Maria José Marcelino. 2014. A review of games designed to improve introductory computer programming competencies. In *2014 IEEE Frontiers in Education Conference (FIE) Proceedings*. 1–7. doi:10.1109/FIE.2014.7044114
- [95] Sepehr Vakil and Maxine McKinney de Royston. 2022. Youth as philosophers of technology. *Mind, Culture, and Activity* 29, 4 (2022), 336–355.
- [96] Jessica Vandenberg, Wookhee Min, Veronica Cateté, Danielle Boulden, and Bradford Mott. 2023. Promoting AI Education for Rural Middle Grades Students with Digital Game Design. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 2 (Toronto ON, Canada) (SIGCSE 2023)*. Association for Computing Machinery, New York, NY, USA, 1388. doi:10.1145/3545947.3576333
- [97] John Vanderhoef. 2013. Casual threats: The feminization of casual video games. (2013).
- [98] Ashok Kumar Veerasamy, Daryl D'Souza, and Mikko-Jussi Laakso. 2016. Identifying novice student programming misconceptions and errors from summative assessments. *Journal of Educational Technology Systems* 45, 1 (2016), 50–73.
- [99] Christiane Gresse Von Wangenheim, Livia S Marques, and Jean CR Hauck. 2020. Machine Learning for All—Introducing Machine Learning in K-12. DOI: <https://doi.org/10.31235/osf.io/wj5ne> (2020).
- [100] Nathan Wang, Paul Tonko, Nikil Ragav, Michael Chungyoun, and Jonathan Plucker. 2023. A perspective on k-12 ai education. *Technology & Innovation* (2023).
- [101] Randi Williams, Hae Won Park, and Cynthia Breazeal. 2019. A is for artificial intelligence: the impact of artificial intelligence activities on young children's perceptions of robots. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–11.
- [102] Gary KW Wong, Xiaojuan Ma, Pierre Dillenbourg, and John Huan. 2020. Broadening artificial intelligence education in K-12: Where to start? *ACM inroads* 11, 1 (2020), 20–29.
- [103] Weipeng Yang. 2022. Artificial Intelligence education for young children: Why, what, and how in curriculum design and implementation. *Computers and Education: Artificial Intelligence* 3 (2022), 100061.
- [104] Iris Heung Yue Yim and Jiahong Su. 2024. Artificial intelligence (AI) learning tools in K-12 education: A scoping review. *Journal of Computers in Education* (2024), 1–39.
- [105] Miao Yue, Morris Siu-Yung Jong, and Yun Dai. 2022. Pedagogical design of K-12 artificial intelligence education: A systematic review. *Sustainability* 14, 23 (2022), 15620.
- [106] Zehui Zhan, Yao Tong, Xixin Lan, and Baichang Zhong. 2024. A systematic literature review of game-based learning in Artificial Intelligence education. *Interactive Learning Environments* 32, 3 (2024), 1137–1158.
- [107] Guoqing Zhou, Chenyang Li, Dianjun Zhang, Dequan Liu, Xiang Zhou, and Jie Zhan. 2021. Overview of Underwater Transmission Characteristics of Oceanic LiDAR. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 14 (2021), 8144–8159. doi:10.1109/JSTARS.2021.3100395